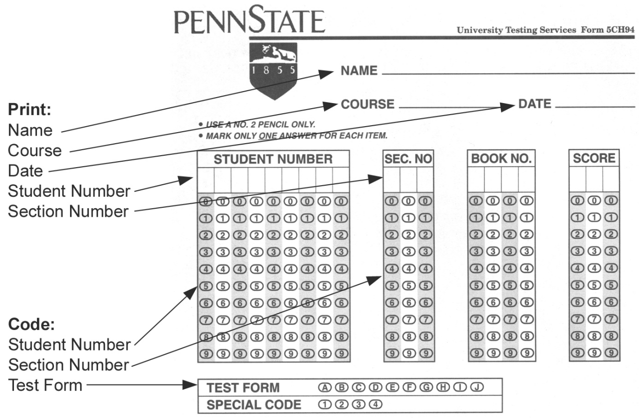
CHEM 112 Timed Practice Exam 3 – Form A

PRINT OUT THIS EXAM TO BRING TO THE TIMED PRACTICE EXAM, BUT *DO NOT COMPLETE IT BEFOREHAND*. BRING A CALCULATOR, and a DATA SHEET AND PERIODIC TABLE; THESE ARE AVAILABLE ON THE EBOOK FOR PRINTING.

Chemistry 112	**Name**	______________________
Timed Practice Exam 3	**Room/Section**	______________________
	Access ID (Email)	___________________

IMPORTANT: On the scantron (answer sheet), you **MUST** clearly fill your **name**, your **student number**, **section number**, and **test form** (white cover = test form A; purple cover = test form B). Use a #2 pencil.



There are 25 4-point questions on this exam. Check that you have done all of the problems and filled in the first 25 bubbles on the scantron. The maximum score on this exam is 100 points. Your score will be reported in percent (maximum of 100%).

Exam policy:

- **Calculators with text-programmable memory are not allowed.**
- Relevant data and formulas, including the periodic table, are attached to this exam.
- Your grade will be based only on what is on the scantron form.
- The answer key will be posted on Canvas after the exam.
- You have 75 minutes to complete this exam.
- **You must completely fill in your scantron bubbles before the 75-minute period has ended.**
- **You must turn in your cover sheet with your scantron answer form.**
- **You must turn in the last question with your handwritten answer by removing the page from this packet and including your name, access ID, and section number.**
- **Turn in your page of personal notes with the last question.**

Advice:

- Fill in your form on the scantron now!
- Underline or circle key words to highlight them for yourself.
- Pay attention to units and magnitudes (decimal places) of numbers obtained from calculations.
- There is no penalty for guessing.
- Circle the answer you choose so that you can grade yourself using the answer key posted after the exam.

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NAME____

CHEM 112 TIMED PRACTICE EXAM 3
Form A: *PUT THIS ON YOUR SCANTRON RIGHT NOW!*

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1. Please rank the following oxides in order from <u>*least to most acidic*</u> in aqueous solution at 25°C:

Cs_2O, SO_3, Sb_2O_5

A. $Cs_2O < SO_3 < Sb_2O_5$
B. $Cs_2O < Sb_2O_5 < SO_3$
C. $SO_3 < Sb_2O_5 < Cs_2O$
D. $SO_3 < Cs_2O < Sb_2O_5$
E. $Sb_2O_5 < SO_3 < Cs_2O$

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2. The auto-ionization constant of water at 90°C is 3.8×10^{-13}. If the pH of an aqueous solution at this temperature is 7.00, what is the pOH?

A. 7.00
B. 7.56
C. 6.44
D. 5.42
E. 12.42

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3. *Proteus mirabilis* is a Gram-negative bacteria responsible for many urinary tract infections (UTIs). Using the enzyme *urease*, the bacteria are able to create an environment with a pH that is easy for them to thrive in. Accordingly, pH testing of urine is frequently performed to assess UTIs. What is the ratio of HPO_4^{2-} to $H_2PO_4^-$ (the primary buffer system of urine) if a patient's urine pH is found to be 9.20 at 25°C?

$K_{a1}(H_3PO_4) = 1.00 \times 10^{-2}$
$K_{a2}(H_3PO_4) = 1.58 \times 10^{-7}$
$K_{a3}(H_3PO_4) = 1.00 \times 10^{-12}$

A. 5.32×10^4
B. 1.58×10^{-3}
C. 2.51×10^2
D. 3.98×10^{-3}
E. 1.58×10^7

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4. Water dissociates in a process called auto-ionization. Consider the following reaction at 90°C:

$$H_2O\ (\ell) \rightleftharpoons H^+(aq) + OH^-(aq),\quad K_{eq} = 3.8 \times 10^{-13}$$

At equilibrium, if $[H^+] = 1 \times 10^{-7}$ M, what is the concentration of hydroxide ions, $[OH^-]$?

A. 1×10^{-7} M
B. 2.8×10^{-8} M
C. 3.6×10^{-7} M
D. 3.8×10^{-6} M
E. 3.8×10^{-13} M

---

5. What are the conjugate acid and conjugate base of the species $HAsO_4^{2-}(aq)$?

	Conjugate Acid	**Conjugate Base**
A.	H_3AsO_4	$H_2AsO_4^-$
B.	AsO_4^{3-}	$H_2AsO_4^-$
C.	$HAsO_4^{2-}$	AsO_4^{3-}
D.	$H_2AsO_4^-$	H_3AsO_4
E.	$H_2AsO_4^-$	AsO_4^{3-}

---

6. Which of the following is the *strongest base* at 25°C in aqueous solution?

A. IO_3^-
B. ClO_4^-
C. ClO_3^-
D. IO_4^-
E. Cl^-

---

7. Which of the following solutions, upon mixing, would create a buffer?

A. 100 mL of 0.1M HCl (aq) + 100 mL of 0.1M NaOH (aq)
B. 100 mL of 0.1M HCl (aq) + 100 mL of 0.1M NH_3 (aq)
C. 100 mL of 0.1M NaOH (aq) + 100 mL of 0.1M CH_3COOH (aq)
D. 200 mL of 0.1M HCl (aq) + 50 mL of 0.1M NH_3 (aq)
E. 50 mL of 0.1M NaOH (aq) + 100 mL of 0.1M CH_3COOH (aq)

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8. Which of the following, when dissolved in water, would create a solution with a pH < 7 at 25°C?

i. $CuCl_2$
ii. $NaBr$
iii. NH_4Cl
iv. CH_3NH_2
v. P_4O_{10}

A. v only
B. ii and iv only
C. iii and v only
D. i, iii, and v only
E. i only

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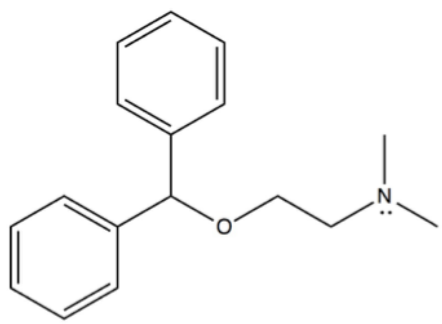
9. Consider an aqueous solution of 1.0 M Benadryl at equilibrium at 25°C. The K_b of Benadryl is 1.0×10^{-5}.

When a scientist adds concentrated HCl to this solution, which of the following would occur?



Structure of Benadryl

i. The pH would increase.
ii. The $[H^+]$ would increase.
iii. The $[OH^-]$ would decrease.
iv. The concentration of Benadryl would decrease.
v. The K_b of Benadryl would decrease.

A. ii and iv only
B. ii, iv, and v only
C. iii and v only
D. i and ii only
E. ii, iii, and iv only

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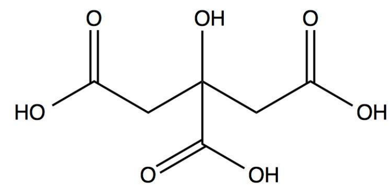
10. Citric acid ($H_3C_6H_5O_7$) is a natural preservative found in citrus fruits. It is also used as an additive to add an acidic or sour taste to foods and beverages. In addition to its many uses, it can also be utilized to create a buffer to study mammalian tissues. A scientist has a 500 mL solution of 1.0 M $H_3C_6H_5O_7$ and 1.0 M $NaH_2C_6H_5O_7$. She then adds in 400 mL of 0.50 M KOH. What is the pH after the addition of KOH at 25°C?

$pK_{a1} = 3.13$, $pK_{a2} = 4.76$, $pK_{a3} = 6.40$



A. 3.50
B. 4.39
C. 5.13
D. 2.76
E. 7.38

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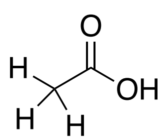
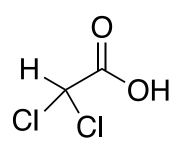
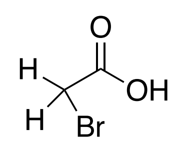
11. Please rank the following carboxylic acids in order from *least to most acidic* in aqueous solution at 25°C:



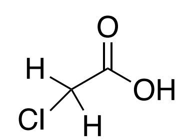
I



II



III



IV

A. III < IV < II < I
B. IV < III < II < I
C. I < III < IV < II
D. I < II < IV < III
E. II < IV < III < I

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12. What is the K_b of methylamine (CH_3NH_2) if ΔG° for this base hydrolysis in water is 19.17 kJ/mol at 25°C?

A. 9.92×10^{-1}
B. 4.36×10^{-4}
C. 8.81×10^{-14}
D. 1.91×10^{-5}
E. 6.78×10^{-7}

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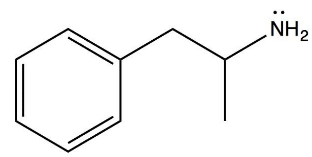
13. A neurologist studying stimulant uptake by neurons adds 25 mL of 1 M HNO_3 to a solution containing 100 mL of 0.5 M amphetamine, a molecule used to treat ADHD and narcolepsy. What is the pH of the solution after the titration at 25°C?
K_b of amphetamine = 6.31×10^{-5}



A. 9.80
B. 4.20
C. 2.25
D. 11.75
E. 7.00

---

14. Which one of the following reactions would have the largest equilibrium constant in water at 25ºC?

A. $HN_3(aq) \rightleftharpoons N_3^-(aq) + H^+(aq)$
B. $HPO_4^{2-}(aq) + H_2O(\ell) \rightleftharpoons H_2PO_4^-(aq) + OH^-(aq)$
C. $CH_3COOH(aq) + KOH(aq) \rightleftharpoons CH_3COOK(aq) + H_2O(\ell)$
D. $NH_3(aq) + H_2O(\ell) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$
E. $HCN(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + CN^-(aq)$

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15. A scientist decides to mix 20 mL of 0.025 M HCl and 5 mL of 0.15 M HNO_3 solutions together for no apparent reason. What is the pH of the resulting solution at 25°C?

A. 0.82
B. 0.76
C. 1.30
D. 2.70
E. 2.90

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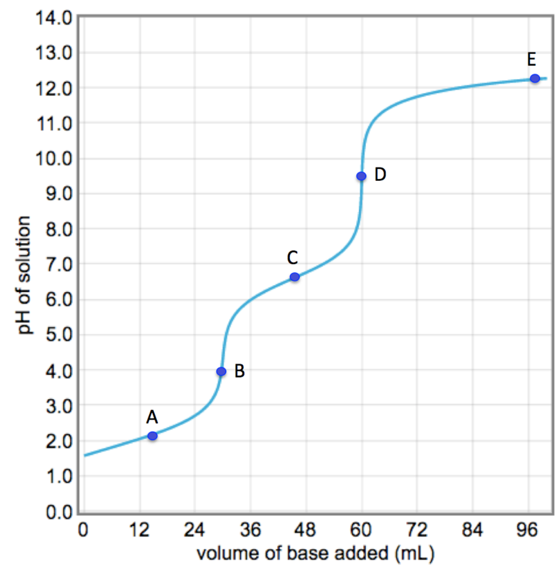
16. Please consider the following titration curve at 25°C. What is the *approximate* **pK_{a2}** of this acid?



A. 2.2
B. 3.9
C. 6.7
D. 9.5
E. 12.4

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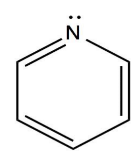
17. For the following reaction, what is the value of Δ*G* at 25°C when the concentration of H_2A is equal to 1.25 M, the concentration of HA^- is equal to 0.75 M, and the pH is equal to 6.10? (For H_2A, $K_{a1} = 2.5 \times 10^{-4}$; $K_{a2} = 4.6 \times 10^{-7}$)

$$H_2A\ (aq) + H_2O\ (\ell) \rightleftharpoons HA^-\ (aq) + H_3O^+\ (aq)$$

A. –18.9 kJ/mol
B. +7.94 kJ/mol
C. +4.76 kJ/mol
D. –6.18 kJ/mol
E. –15.5 kJ/mol

---

18. Pyridine (C_5H_5N, structure below) is a widely used building block and chemical reagent in the synthesis of organic compounds, including pharmaceuticals. What is the K_b of pyridine if a 1.5 M solution has a pH of 9.71 at 25°C?



A. 2.53×10^{-20}
B. 5.12×10^{-5}
C. 1.95×10^{-10}
D. 1.75×10^{-9}
E. 2.13×10^{-12}

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19. Cyanide acts as a potent inhibitor of respiration. Being highly toxic, sodium cyanide (NaCN) is an illegal method of collecting live fish by stunning them with low concentrations of the salt. A wayward scientist finds a new species of fish in the waters of Southeast Asia and decides to use a 0.65 M NaCN solution to stun the fish and bring it back to his lab alive for further study. What is the pH of the aqueous NaCN solution at 25°C?
K_a of HCN = 6.17×10^{-10}

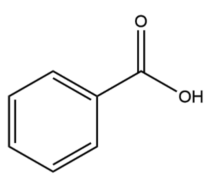
A. 4.70
B. 9.30
C. 11.51
D. 2.49
E. 9.21

---

20. Which of the following acts as a Lewis acid?
A. NH_3
B. $AlCl_3$
C. CH_3CH_2OH
D. H_2
E. Cl^-

---

21. What is the percent ionization of a 5.0 M solution of benzoic acid ($K_a = 6.46 \times 10^{-5}$) at 25°C?



Structure of benzoic acid

A. 0.80 %
B. 1.80 %
C. 0.36 %
D. 1.21 %
E. 0.16 %

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22. Which one of the following statements is TRUE concerning a solution of 0.1M H_3PO_4 at 25°C?
$K_{a1}(H_3PO_4) = 1.00 \times 10^{-2}$, $K_{a2}(H_3PO_4) = 1.58 \times 10^{-7}$, $K_{a3}(H_3PO_4) = 1.00 \times 10^{-12}$

A. The $[H^+]$ at equilibrium is calculated from K_{a3}.
B. PO_4^{3-} (aq) is a weaker base than HPO_4^{2-} (aq).
C. $H_2PO_4^-$ (aq) is a stronger acid than HPO_4^{2-} (aq).
D. H_3PO_4 (aq) can best be classified as an amphoteric species.
E. The pH of the 0.1 M solution of H_3PO_4 is 1.0.

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23. A titration enthusiast titrates a 100 mL solution of 1.0 M CH_3COOH with 0.50 M NaOH. What is the pH **at the equivalence point** of this titration at 25°C? (K_a of $CH_3COOH = 1.8 \times 10^{-5}$)

A. 4.69
B. 11.39
C. 2.61
D. 9.13
E. 8.87

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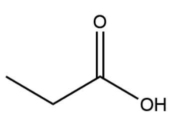
24. Propionic acid, CH_3CH_2COOH, is found in both Swiss cheese and sweat. What is the pH of a 0.25 M solution of propionic acid ($K_a = 1.34 \times 10^{-5}$) at 25°C?



Structure of propionic acid

A. 7.00
B. 4.87
C. 5.47
D. 2.74
E. 4.43

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The last question on this exam requires a handwritten answer. You must answer the question in the space below, detach this page from the exam packet, write your name, student number, Penn State email and section on it, and turn it in at the exam or you will not get credit.

NAME (Readable, 0.2 points)____

STUDENT NUMBER______________________________________

PSU ACCESS ID EMAIL (such as abc12@psu.edu) ________________________________

SECTION _________

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25. Handwritten Answer Question: Make your writing **readable** or no points will be awarded. (3.8 points).

Please place the following compounds in the table under the correct classification:

$HONH_2$

HOI

ClO_3^-

N_3^-

Strong Acid	Strong Base	Weak Acid	Weak Base	Spectator	Non-electrolyte

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The End – You did it!! Great work!

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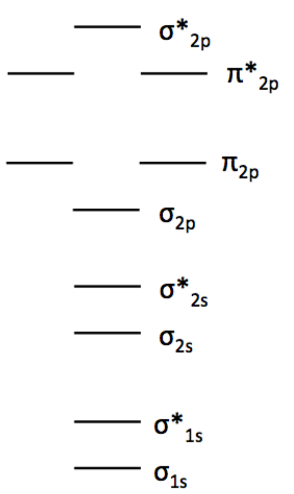
CHEM 112 Constants, Conversions and Equations

Conversions	Constants	
$1\,\text{J} = \text{kg}\cdot\text{m}^2\cdot\text{s}^{-2} = \text{C}\cdot\text{V}$	$c = 2.99792458\times10^{8}\ \text{m}\cdot\text{s}^{-1}$	$R = 0.08206\ \text{L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
$1\,\text{A} = \text{C}\cdot\text{s}^{-1}$	$h = 6.63\times10^{-34}\ \text{J}\cdot\text{s}$	$R = 8.314\ \text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
$1\,\text{g} = 6.02\times10^{23}$ amu	$F = 96{,}485\ \text{C}\cdot\text{mol}^{-1}$	${}^{1}_{0}\text{n} = 1.0086649$ amu
	$N_A = 6.022\times10^{23}\ \text{mol}^{-1}$	${}^{1}_{1}\text{p} = 1.0072765$ amu
	$K_W = 1\times10^{-14}$ at 25 °C	

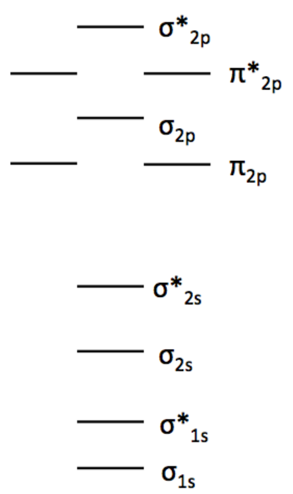
Equations

$k = Ae^{\left(\frac{-E_a}{RT}\right)}$	$\text{Rate} = kN$	$\Delta G = \Delta H - T\Delta S = -nFE$
$\ln k = \frac{-E_a}{RT} + \ln A$	$\ln\frac{N_t}{N_0} = \frac{-0.693t}{t_{1/2}} = -kt$	$\Delta G^o = -2.303\,RT\log K = -RT\ln K$
$\ln[A] = \ln[A]_0 - kt$	$\text{E} = h\nu = \frac{hc}{\lambda}$	$\Delta G = \Delta G^o + RT\ln Q$
$\ln\frac{k_1}{k_2} = \frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$	$E = mc^2$	$\Delta G = \Delta G^o + 2.303\,RT\log Q$
$t_{1/2} = \frac{0.693}{k}$ $\quad t_{1/2} = \frac{1}{k[A]_0}$	$K_a \times K_b = K_W = [\text{H}^+][\text{OH}^-]$	$E^o = \frac{0.0592}{n}\log K$ (at 25 °C)
$\frac{1}{[A]} = \frac{1}{[A]_0} + \text{kt}$	$\text{pH} = \text{p}K_a + \log\frac{[\text{X}^-]}{[\text{HX}]}$	$E = E^o - \frac{0.0592}{n}\log Q$ (at 25 °C)
$K_p = K_c(RT)^{\Delta n}$	$\text{p}K_W = \text{pH} + \text{pOH} = \text{p}K_a + \text{p}K_b$	

Molecular Orbital Theory Diagrams

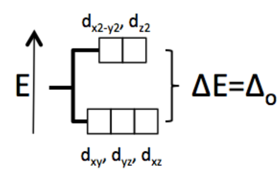


Small 2s-2p Interaction (O_2, F_2, Ne_2)

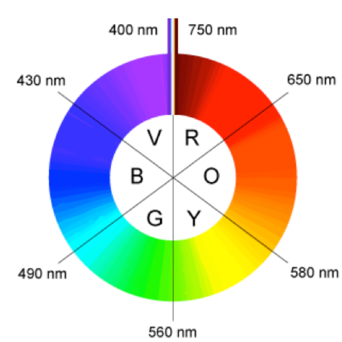


Large 2s-2p Interaction (B_2, C_2, N_2)

Crystal Field Theory Diagram



Color Wheel with Wavelength Ranges



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Standard Reduction Potentials in Water at 25°C Half Reaction	$E^\circ_{1/2}$(V)
$F_2(g) + 2\,e^- \rightarrow 2\,F^-(aq)$	+2.87
$H_2O_2(aq) + 2\,H^+(aq) + 2\,e^- \rightarrow 2\,H_2O(\ell)$	+1.776
$Au^+(aq) + e^- \rightarrow Au(s)$	+1.69
$PbO_2(s) + 4\,H^+(aq) + SO_4^{2-}(aq) + 2\,e^- \rightarrow PbSO_4(s) + 2\,H_2O(\ell)$	+1.685
$Cl_2(g) + 2\,e^- \rightarrow 2\,Cl^-(aq)$	+1.359
$O_2(g) + 4\,H^+(aq) + 4\,e^- \rightarrow 2\,H_2O(\ell)$	+1.23
$ClO_4^-(aq) + 2\,H^+(aq) + 2\,e^- \rightarrow ClO_3^-(aq) + H_2O(\ell)$	+1.23
$Pt^{2+}(aq) + 2\,e^- \rightarrow Pt(s)$	+1.20
$2\,IO_3^-(aq) + 12\,H^+(aq) + 10\,e^- \rightarrow I_2(s) + 6\,H_2O(\ell)$	+1.195
$Br_2(\ell) + 2\,e^- \rightarrow 2\,Br^-(aq)$	+1.065
$NO_3^-(aq) + 4\,H^+(aq) + 3\,e^- \rightarrow NO(g) + 2\,H_2O(\ell)$	+0.96
$Ag^+(aq) + e^- \rightarrow Ag(s)$	+0.799
$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	+0.771
$O_2(g) + 2\,H^+(aq) + 2\,e^- \rightarrow H_2O_2(aq)$	+0.68
$I_2(s) + 2\,e^- \rightarrow 2\,I^-(aq)$	+0.536
$I_3^-(aq) + 2\,e^- \rightarrow 3\,I^-(aq)$	+0.53
$Cu^+(aq) + e^- \rightarrow Cu(s)$	+0.521
$Cu^{2+}(aq) + 2\,e^- \rightarrow Cu(s)$	+0.337
$Sn^{4+}(aq) + 2\,e^- \rightarrow Sn^{2+}(aq)$	+0.154
$Cu^{2+}(aq) + e^- \rightarrow Cu^+(aq)$	+0.153
$2\,H^+(aq) + 2\,e^- \rightarrow H_2(g)$	+0.00
$Pb^{2+}(aq) + 2\,e^- \rightarrow Pb(s)$	−0.126
$Sn^{2+}(aq) + 2\,e^- \rightarrow Sn(s)$	−0.136
$Co^{2+}(aq) + 2\,e^- \rightarrow Co(s)$	−0.277
$Ni^{2+}(aq) + 2\,e^- \rightarrow Ni(s)$	−0.28
$Cd^{2+}(aq) + 2\,e^- \rightarrow Cd(s)$	−0.403
$Fe^{2+}(aq) + 2\,e^- \rightarrow Fe(s)$	−0.440
$Cr^{3+}(aq) + 3\,e^- \rightarrow Cr(s)$	−0.74
$Zn^{2+}(aq) + 2\,e^- \rightarrow Zn(s)$	−0.763
$2\,H_2O(\ell) + 2\,e^- \rightarrow H_2(g) + 2\,OH^-(aq)$	−0.83
$SO_4^{2-}(aq) + H_2O(\ell) + 2\,e^- \rightarrow SO_3^{2-}(aq) + 2\,OH^-(aq)$	−0.93
$Mn^{2+}(aq) + 2\,e^- \rightarrow Mn(s)$	−1.18
$Al^{3+}(aq) + 3\,e^- \rightarrow Al(s)$	−1.66
$Mg^{2+}(aq) + 2\,e^- \rightarrow Mg(s)$	−2.37
$Na^+(aq) + e^- \rightarrow Na(s)$	−2.71
$Ca^{2+}(aq) + 2\,e^- \rightarrow Ca(s)$	−2.87
$Li^+(aq) + e^- \rightarrow Li(s)$	−3.05

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PERIODIC TABLE of the ELEMENTS

MAIN GROUPS | TRANSITION METALS | MAIN GROUPS

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 **H** 1.008																1 **H** 1.008	2 **He** 4.003
3 **Li** 6.941	4 **Be** 9.012											5 **B** 10.811	6 **C** 12.011	7 **N** 14.007	8 **O** 15.999	9 **F** 18.998	10 **Ne** 20.180
11 **Na** 22.990	12 **Mg** 24.305											13 **Al** 26.982	14 **Si** 28.086	15 **P** 30.974	16 **S** 32.066	17 **Cl** 35.453	18 **Ar** 39.948
19 **K** 39.098	20 **Ca** 40.078	21 **Sc** 44.956	22 **Ti** 47.867	23 **V** 50.942	24 **Cr** 51.996	25 **Mn** 54.938	26 **Fe** 55.845	27 **Co** 58.933	28 **Ni** 58.693	29 **Cu** 63.546	30 **Zn** 65.39	31 **Ga** 69.723	32 **Ge** 72.61	33 **As** 74.992	34 **Se** 78.96	35 **Br** 79.904	36 **Kr** 83.80
37 **Rb** 85.468	38 **Sr** 87.62	39 **Y** 88.906	40 **Zr** 91.224	41 **Nb** 92.906	42 **Mo** 95.94	43 **Tc** [98]	44 **Ru** 101.07	45 **Rh** 102.90	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.76	52 **Te** 127.60	53 **I** 126.90	54 **Xe** 131.29
55 **Cs** 132.91	56 **Ba** 137.33	57 **La*** 138.91	72 **Hf** 178.49	73 **Ta** 180.95	74 **W** 183.84	75 **Re** 186.21	76 **Os** 190.23	77 **Ir** 192.22	78 **Pt** 195.08	79 **Au** 196.97	80 **Hg** 200.59	81 **Tl** 204.38	82 **Pb** 207.2	83 **Bi** 208.98	84 **Po** [209]	85 **At** [210]	86 **Rn** [222]
87 **Fr** [223]	88 **Ra** [226]	89 **Ac**** [227]	104 **Rf** [261]	105 **Db** [262]	106 **Sg** [266]	107 **Bh** [264]	108 **Hs** [265]	109 **Mt** [268]	110 **Ds** [271]	111 **Rg** [280]	112 **Cn** [285]	113 **Nh** [286]	114 **Fl** [289]	115 **Mc** [289]	116 **Lv** [293]	117 **Ts** [294]	118 **Og** [294]

* LANTHANOIDS	58 **Ce** 140.12	59 **Pr** 140.91	60 **Nd** 144.24	61 **Pm** [145]	62 **Sm** 150.36	63 **Eu** 151.96	64 **Gd** 157.25	65 **Tb** 158.92	66 **Dy** 162.50	67 **Ho** 164.93	68 **Er** 167.26	69 **Tm** 168.93	70 **Yb** 173.04	71 **Lu** 174.97
** ACTINOIDS	90 **Th** 232.04	91 **Pa** 231.04	92 **U** 238.03	93 **Np** [237]	94 **Pu** [244]	95 **Am** [243]	96 **Cm** [247]	97 **Bk** [247]	98 **Cf** [251]	99 **Es** [252]	100 **Fm** [257]	101 **Md** [258]	102 **No** [259]	103 **Lr** [262]